1)

```
Main:
        ADD X2, XZR, #100      ; X2 = 100
        ADD X0, XZR, #5        ; let n = 5 can be any value
        BL SUM
SUM:    SUB X2, X2, X1         ; 100 - Σn
SUM:    SUBI SP, SP #16
        STUR LR, [SP, #8]
        STUR X0, [SP, #0]
        SUBIS XZR, X0 #1       Compare n & 1
        B.GE L1                if n ≥ 1, go to L1.
        ADDI X1, XZR, #0       else set return
        ADDI SP, SP, #16.      Pop stack
        BR LR                  return
L1:     SUBI X0, X0, #1        n = n - 1
        BL SUM                 call Sum(n-1)
        LDUR X0, [SP, #0]      restore caller n
        LDUR LR, [SP, #8]      restore after return
        ADD X1, X0, X1
        BR LR
```

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2) 11111000000 | 01111 0000 | 00 | 01010 | 01001

STUR X9, [X10, #240]

3) LDUR X9, [X10, #16]

D-format

opcode : $7C_2$ in hex : 111 1100 0010

offset : 16_{10} : 0000010000

OP2 : : 00

Rn : 10 : 01010

Rt : 9 : 01001

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SUB x9, x19, x9

R-Format

opcode: 658 : 110 0101 1000

Rm [5bit] : 01001 [9]

Shamt [6bits] : 000 000

Rm [5bits] : 10011 [19]

Rd [5bits] = 01001 [9]

1 1 0 0 1 0 1 1 0 0 0 1 0 0 1 0 0 1 0 0 0 0 0 0

1 0 0 1 1 0 1 0 0 1

STUR x9, [x10, #32]

D-format

opcode : 0x7C0 = 111 1100 0000

offset : 3210 = 00100000

OP2 : 00

Rn : 01010 [10]

Rt : 01001 [9]

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CBZ, x9, LOOP

CB-format.

OPcode: 5A8 to 5AF : 101 1010 1xxx

: 101 1010 1

offset: 3 inst to skip backward: -3

111 1111 1111 1111 1101

Rt = 01001 [9]

101 1010 1111 1111 1111 1111

1101 01001

4) For given fact, each time 2 words are pushed onto stack

For x0 = 4, [4, 3, 2, 1, 0]

Total number of words = 5 × 2 = 10 words

5)

ADDI x9, XZR #20	0 - - - 000010100
SUBI x10, XZR, #5	1 - - - 111111011
LSL x11, x9 #3	0 - - - 100110000
ORR x11, x11, x10	1 - - - 111111011

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